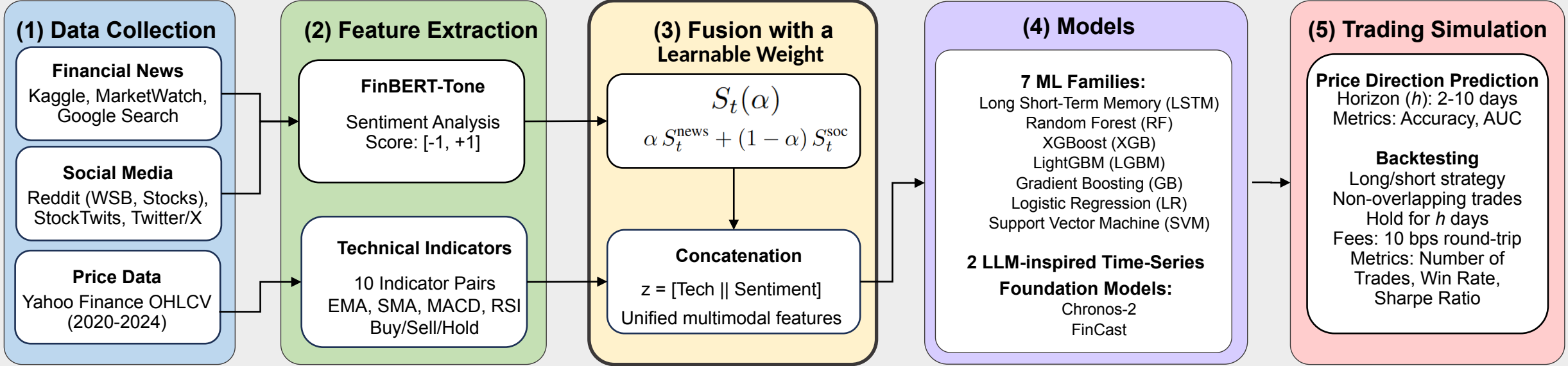
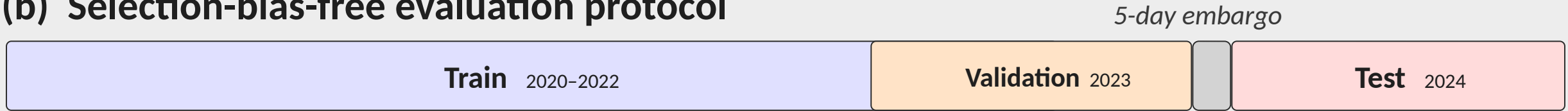


(a) Multimodal pipeline under evaluation



(b) Selection-bias-free evaluation protocol



- Note:
- Fit on training; tune and **select model family + horizon on the validation window**; the choice is locked to an immutable file (selected_config.json).
 - Test year **reported once** — no re-selection.

Grid: 8 feature sets $\times$ 7 models $\times$ 5 stocks $\times$ 3 horizons $\times$ 5 seeds = **4,200 task-specific runs** (plus 2 time-series foundation models).
Reported as multi-seed mean $\pm$ std with paired Wilcoxon signed-rank tests.